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Windows PowerShell
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PS C:\WINDOWS\system32> wsl
sampath@MS-Desktop:/mnt/c/WINDOWS/system32$ cd o s
-bash: cd: too many arguments
sampath@MS-Desktop:/mnt/c/WINDOWS/system32$ cd os
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os$ cd skill
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ nano skill4.c
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ gcc skill4.c -o skill4
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls -l

===== TOKEN STREAM =====
Token 1: ls           Type: COMMAND
Token 2: -l           Type: ARGUMENT

Token validation successful.

===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: ls
+---- ARGUMENT: -l

===== EXECUTION STRUCTURE =====
Command: ls
Arguments: -l
Background Execution: NO

=====
Parsing completed successfully.
=====
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls -l /home

===== TOKEN STREAM =====
Token 1: ls           Type: COMMAND
Token 2: -l           Type: ARGUMENT
Token 3: /home        Type: ARGUMENT

Token validation successful.

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===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: ls
+---- ARGUMENT: -l
+---- ARGUMENT: /home

===== EXECUTION STRUCTURE =====
Command: ls
Arguments: -l /home
Background Execution: NO

=====
Parsing completed successfully.
=====
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill1$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls -l | grep txt

===== TOKEN STREAM =====
Token 1: ls           Type: COMMAND
Token 2: -l           Type: ARGUMENT
Token 3: |             Type: PIPE
Token 4: grep         Type: ARGUMENT
Token 5: txt          Type: ARGUMENT

Token validation successful.

===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: ls
+---- ARGUMENT: -l
|
+---- PIPE
|
+---- ARGUMENT: grep
+---- ARGUMENT: txt

Execution Structure:
Command: ls
Argument: -l

Pipe detected -> Next command

===== EXECUTION STRUCTURE =====
Command: grep
Arguments: txt
Background Execution: NO

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=====
Parsing completed successfully.
=====
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls -l > output.txt

===== TOKEN STREAM =====
Token 1: ls           Type: COMMAND
Token 2: -l           Type: ARGUMENT
Token 3: >            Type: REDIRECT_OUT
Token 4: output.txt   Type: ARGUMENT

Token validation successful.

===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: ls
+---- ARGUMENT: -l
+---- OUTPUT_REDIRECTION
|       File: output.txt

===== EXECUTION STRUCTURE =====
Command: ls
Arguments: -l
Output File: output.txt
Background Execution: NO

=====
Parsing completed successfully.
=====
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> cat < input.txt^[[C

===== TOKEN STREAM =====
Token 1: cat           Type: COMMAND
Token 2: <             Type: REDIRECT_IN
Token 3: input.txt     Type: ARGUMENT

Token validation successful.

===== PARSE TREE =====

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sampath@MS-Desktop: /mnt/c/WINDOWS/system32/os/skill

===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: cat
+---- INPUT_REDIRECTION
|      File: input.txt

===== EXECUTION STRUCTURE =====
Command: cat
Arguments:
Input File: input.txt
Background Execution: NO

=====
Parsing completed successfully.
=====
sampath@MS-Desktop: /mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls >> output.txt

===== TOKEN STREAM =====
Token 1: ls           Type: COMMAND
Token 2: >>           Type: APPEND
Token 3: output.txt   Type: ARGUMENT

Token validation successful.

===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: ls
+---- APPEND_REDIRECTION
|      File: output.txt

===== EXECUTION STRUCTURE =====
Command: ls
Arguments:
Output File: output.txt (APPEND)
Background Execution: NO

=====
Parsing completed successfully.
=====
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=====
Parsing completed successfully.
=====
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> gedit &

===== TOKEN STREAM =====
Token 1: gedit          Type: COMMAND
Token 2: &              Type: BACKGROUND

Token validation successful.

===== PARSE TREE =====
COMMAND_LINE
+---- COMMAND: gedit
+---- ARGUMENT: &

===== EXECUTION STRUCTURE =====
Command: gedit
Arguments:
Background Execution: YES

=====
Parsing completed successfully.
=====
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> | ls

===== TOKEN STREAM =====
Token 1: |              Type: PIPE
Token 2: ls             Type: ARGUMENT

Syntax Error: Command cannot start with '|'.
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls |

===== TOKEN STREAM =====

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Token 1: ls      Type: COMMAND
Token 2: |       Type: PIPE
Syntax Error: Command cannot start with '|'.
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls |

===== TOKEN STREAM =====
Token 1: ls      Type: COMMAND
Token 2: |       Type: PIPE
Syntax Error: Command cannot end with '|'.
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
> ls >

===== TOKEN STREAM =====
Token 1: ls      Type: COMMAND
Token 2: >       Type: REDIRECT_OUT
Syntax Error: Redirection requires a filename.
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ ./skill4
=====
SIMPLE SHELL TOKENIZER & PARSER
=====

Enter a command:
>

===== TOKEN STREAM =====
No tokens found.

Empty command.
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/skill$ _

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